RoFormer: Enhanced Transformer with Rotary Position Embedding

Department of Computer Science & Artificial Intelligence

Abstract

This paper investigates architectures with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding RoFormer: Enhanced Transformer with Rotary Position Embedding.

2. Technical Methodology & Empirical Findings

Section topic: Architectures
Question context: What is Rotary Position Embedding (RoPE) and how does it encode relative token distances?
Key Finding: RoPE applies complex orthogonal rotation matrices to query and key vectors such that their inner product directly incorporates relative positional distance [RoFormer, p. 5].
Evidence passage: Multiplies key and query representations by orthogonal rotation matrices Inner product naturally decays with increasing token distance

3. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for RoFormer: Enhanced Transformer with Rotary Position Embedding. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

4. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for RoFormer: Enhanced Transformer with Rotary Position Embedding. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.